

Introduction to Optimization: Part 2

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The tables and chairs problem, revisited

$$\begin{array}{ll}\max & 20x_1 + 30x_2 \\ \text{such that} & x_1 + 2x_2 \leq 30 \quad (\text{Labor}) \\ & 10x_1 + 5x_2 \leq 120 \quad (\text{Wood}) \\ & x_1, x_2 \geq 0\end{array}$$

Optimal solution was $\mathbf{x}^* = (x_1^*, x_2^*) = (6, 12)$, $\Rightarrow$ We could solve the LP again with the new values...
with a optimal objective value of 480 (maximum profit \$480).

- What is the effect of 1 more labor hour / 1 more unit of wood on profits?
 - What happens if chairs yielded \$25 profit instead? Would the current production plan still be optimal?
- But what if there was a simpler way?

The not-so-simple way: just plug in and resolve

One more unit of labor:

$$\begin{array}{ll}\max & 20x_1 + 30x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq \mathbf{31} \quad (\text{Labor}) \\ & 10x_1 + 5x_2 \leq 120 \quad (\text{Wood}) \\ & x_1, x_2 \geq 0\end{array}$$

Optimal solution is now $\mathbf{x^* = (\frac{17}{3}, \frac{38}{3})}$,
with a optimal objective value of $493\frac{1}{3}$.

One more unit of wood:

$$\begin{array}{ll}\max & 20x_1 + 30x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 30 \quad (\text{Labor}) \\ & 10x_1 + 5x_2 \leq \mathbf{121} \quad (\text{Wood}) \\ & x_1, x_2 \geq 0\end{array}$$

Optimal solution is now $\mathbf{x^* = (\frac{92}{15}, \frac{179}{15})}$,
with a optimal objective value of $480\frac{2}{3}$.

The “simpler” way involves the dual problem

- Every linear program has a dual, which is also a linear program
- Variables in the primal LP correspond to constraints in the dual LP
- Constraints in the primal LP correspond to variables in the dual LP
- A maximization in the primal LP becomes a minimization in the dual LP, and vice versa
- The dual of the dual is the primal
- Their optimal objective values are the same!
(when they both have an optimal solution)

Deriving the dual problem

Let's start by writing the problem in compact form:

$$\max \{ \mathbf{c}^\top \mathbf{x} \mid \mathbf{Ax} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0} \}$$

1. Let's relax the constraints $\mathbf{Ax} \leq \mathbf{b}$, and add a penalty to the objective based on a penalty vector $\mathbf{p}$:

$$\begin{aligned} Z^* = \max_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

$$\begin{aligned} g(\mathbf{p}) = \max_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} + \mathbf{p}^\top (\mathbf{b} - \mathbf{Ax}) \\ \text{s.t.} \quad & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

- One can write the objective of the relaxed problem as:

$$\mathbf{c}^\top \mathbf{x} + \sum_{i \in [n]} p_i (b_i - \mathbf{a}_i^\top \mathbf{x})$$

This way, p_i is the "penalty" for the i -th constraint $\mathbf{a}_i^\top \mathbf{x} \leq b_i$ (more on this later!)

Deriving the dual problem (cont.)

2. $g(\mathbf{p})$ is an upper bound on the true optimal objective Z^* ,
(for whatever $\mathbf{p}$ you use that is $\geq \mathbf{0}$)

$$\begin{aligned} g(\mathbf{p}) &\geq \mathbf{c}^\top \mathbf{x}^* + \underbrace{\mathbf{p}^\top}_{\geq 0} \underbrace{(\mathbf{b} - \mathbf{A}\mathbf{x}^*)}_{\geq 0} \\ &\geq \mathbf{c}^\top \mathbf{x}^* \end{aligned}$$

- The first inequality is because $g(\mathbf{p})$ is the optimal objective for the relaxed LP, and $\mathbf{x}^*$ is feasible for the relaxed LP
- The second inequality is because $\mathbf{b} - \mathbf{A}\mathbf{x}^* \geq \mathbf{0}$ (since $\mathbf{x}^*$ is feasible for the original LP), and $\mathbf{p} \geq \mathbf{0}$

Deriving the dual problem (cont.)

3. Find the *tightest* upper bound: $\min_{\mathbf{p}} g(\mathbf{p})$

$$\begin{aligned} & \min \left\{ g(\mathbf{p}) \mid \mathbf{p} \geq \mathbf{0} \right\} \\ &= \min \left\{ \max \left\{ \mathbf{c}^\top \mathbf{x} + \mathbf{p}^\top \mathbf{b} - \mathbf{p}^\top \mathbf{A} \mathbf{x} \mid \mathbf{x} \geq \mathbf{0} \right\} \mid \mathbf{p} \geq \mathbf{0} \right\} \\ &= \min \left\{ \mathbf{p}^\top \mathbf{b} + \max \left\{ (\mathbf{c}^\top - \mathbf{p}^\top \mathbf{A}) \mathbf{x} \mid \mathbf{x} \geq \mathbf{0} \right\} \mid \mathbf{p} \geq \mathbf{0} \right\} \end{aligned}$$

- The inner maximization is:

$$\max \left\{ (\mathbf{c}^\top - \mathbf{p}^\top \mathbf{A}) \mathbf{x} \mid \mathbf{x} \geq \mathbf{0} \right\} = \begin{cases} 0 & \text{if } \mathbf{c}^\top - \mathbf{p}^\top \mathbf{A} \leq \mathbf{0} \\ \infty & \text{otherwise.} \end{cases}$$

- This means:

$$\min \left\{ g(\mathbf{p}) \mid \mathbf{p} \geq \mathbf{0} \right\} = \min \left\{ \mathbf{p}^\top \mathbf{b} \mid \mathbf{p}^\top \mathbf{A} \geq \mathbf{c}^\top, \mathbf{p} \geq \mathbf{0} \right\}$$

Deriving the dual problem (cont.)

Primal problem

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

Dual problem

$$\begin{aligned} \min \quad & \mathbf{p}^\top \mathbf{b} \\ \text{s.t.} \quad & \mathbf{p}^\top \mathbf{A} \geq \mathbf{c}^\top \\ & \mathbf{p} \geq \mathbf{0} \end{aligned}$$

- One dual variable p_j per constraint $j \in \{1, \dots, m\}$
- Primal cost vector $\mathbf{c} \rightarrow$ right-hand-side of dual constraints
- RHS of primal constraints $\mathbf{b} \rightarrow$ dual cost vector
- Max becomes min and vice versa
- Free variables in primal induce equality constraints in the dual, and signed variables in primal induce inequality constraints in the dual (and vice versa)

Taking the dual of the tables and chairs problem: step by step

$$\max \quad 20x_1 + 30x_2$$

$$\text{s.t.} \quad x_1 + 2x_2 \leq 30 \quad [p_1]$$

$$10x_1 + 5x_2 \leq 120 \quad [p_2]$$

$$x_1, x_2 \geq 0$$

$$\min \quad 30p_1 + 120p_2$$

$$\text{s.t.} \quad p_1 + 10p_2 \geq 20 \quad [x_1]$$

$$2p_1 + 5p_2 \geq 30 \quad [x_2]$$

$$p_1 \geq 0$$

$$p_2 \geq 0$$

- Start by assigning dual variables for each constraint:
e.g. p_1 for labor, and p_2 for wood
- Write down the dual objective
- Write down the dual constraint for primal variable x_1
- Write down the dual constraint for primal variable x_2
- Write the sign constraints for the dual variables

Derive the dual of the standard-form LP: $\min \{ \mathbf{c}^\top \mathbf{x} \mid \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \}$.

Deriving the dual of the standard-form LP

Let's start with a problem in standard form:

$$\min \{ \mathbf{c}^\top \mathbf{x} \mid \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \}$$

1. Let's relax the constraints $\mathbf{Ax} = \mathbf{b}$, and add a a penalty to the objective based on a penalty vector $\mathbf{p}$:

$$Z^* = \min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x}$$

$$\text{s.t. } \mathbf{Ax} = \mathbf{b}$$

$$\mathbf{x} \geq \mathbf{0}$$

$$g(\mathbf{p}) = \min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x} + \mathbf{p}^\top (\mathbf{b} - \mathbf{Ax})$$

$$\text{s.t. } \mathbf{x} \geq \mathbf{0}$$

- One can write the objective of the relaxed problem as:

$$\mathbf{c}^\top \mathbf{x} + \sum_{i \in [n]} p_i (b_i - \mathbf{a}_i^\top \mathbf{x})$$

This way, p_i is the "penalty" for the i -th constraint $\mathbf{a}_i^\top \mathbf{x} = b_i$

Deriving the dual problem (cont.)

2. $g(\mathbf{p})$ is a lower bound on the true optimal objective Z^* ,
(for whatever $\mathbf{p}$ you use)

$$\begin{aligned} g(\mathbf{p}) &\leq \mathbf{c}^\top \mathbf{x}^* + \mathbf{p}^\top \underbrace{(\mathbf{b} - \mathbf{A}\mathbf{x}^*)}_{=0} \\ &= \mathbf{c}^\top \mathbf{x}^* \end{aligned}$$

- The first inequality is because $g(\mathbf{p})$ is the optimal objective for the relaxed LP, and $\mathbf{x}^*$ is feasible for the relaxed LP
- The second inequality is because $\mathbf{b} - \mathbf{A}\mathbf{x}^* = \mathbf{0}$, since $\mathbf{x}^*$ is feasible for the original LP

Deriving the dual of the standard-form LP (cont.)

3. Find the *tightest* lower bound: $\max_{\mathbf{p}} g(\mathbf{p})$

$$\begin{aligned} & \max \left\{ g(\mathbf{p}) \mid \mathbf{p} \in \mathbb{R}^m \right\} \\ &= \max \left\{ \min \left\{ \mathbf{c}^\top \mathbf{x} + \mathbf{p}^\top \mathbf{b} - \mathbf{p}^\top \mathbf{A} \mathbf{x} \mid \mathbf{x} \geq \mathbf{0} \right\} \mid \mathbf{p} \in \mathbb{R}^m \right\} \\ &= \max \left\{ \mathbf{p}^\top \mathbf{b} + \min \left\{ (\mathbf{c}^\top - \mathbf{p}^\top \mathbf{A}) \mathbf{x} \mid \mathbf{x} \geq \mathbf{0} \right\} \mid \mathbf{p} \in \mathbb{R}^m \right\} \end{aligned}$$

- The inner minimization is:

$$\min \left\{ (\mathbf{c}^\top - \mathbf{p}^\top \mathbf{A}) \mathbf{x} \mid \mathbf{x} \geq \mathbf{0} \right\} = \begin{cases} 0 & \text{if } \mathbf{c}^\top - \mathbf{p}^\top \mathbf{A} \geq \mathbf{0} \\ -\infty & \text{otherwise.} \end{cases}$$

- This means:

$$\max \left\{ g(\mathbf{p}) \mid \mathbf{p} \in \mathbb{R}^m \right\} = \max \left\{ \mathbf{p}^\top \mathbf{b} \mid \mathbf{p}^\top \mathbf{A} \leq \mathbf{c}^\top, \mathbf{p} \in \mathbb{R}^m \right\}$$

The dual of a standard-form LP

Primal problem

$$\begin{array}{ll}\min & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

Dual problem

$$\begin{array}{ll}\max & \mathbf{p}^\top \mathbf{b} \\ \text{s.t.} & \mathbf{p}^\top \mathbf{A} \leq \mathbf{c}^\top \\ & \mathbf{p} \in \mathbb{R}^m\end{array}$$

- Again, we see that primal constraints relate to dual variables, while primal variables relate to dual constraints
- The signs of primal constraints affect the signs of dual variables, and the signs of primal variables affect the signs of dual constraints

The general form of the LP dual

Here's how the LP dual can be formed from a general linear optimization problem with arbitrary sign constraints:

Primal problem

$$\begin{array}{ll}\max & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{a}_i^\top \mathbf{x} \geq b_i, \quad \forall i \in \mathcal{M}_1 \\ & \mathbf{a}_i^\top \mathbf{x} \leq b_i, \quad \forall i \in \mathcal{M}_2 \\ & \mathbf{a}_i^\top \mathbf{x} = b_i, \quad \forall i \in \mathcal{M}_0 \\ & x_j \geq 0, \quad \forall j \in \mathcal{N}_1 \\ & x_j \leq 0, \quad \forall j \in \mathcal{N}_2 \\ & x_j \text{ free}, \quad \forall j \in \mathcal{N}_0\end{array}$$

Dual problem

$$\begin{array}{ll}\min & \mathbf{p}^\top \mathbf{b} \\ \text{s.t.} & p_i \leq 0, \quad \forall i \in \mathcal{M}_1 \\ & p_i \geq 0, \quad \forall i \in \mathcal{M}_2 \\ & p_i \text{ free}, \quad \forall i \in \mathcal{M}_0 \\ & \mathbf{p}^\top \mathbf{A}_j \geq c_j, \quad \forall j \in \mathcal{N}_1 \\ & \mathbf{p}^\top \mathbf{A}_j \leq c_j, \quad \forall j \in \mathcal{N}_2 \\ & \mathbf{p}^\top \mathbf{A}_j = c_j, \quad \forall j \in \mathcal{N}_0\end{array}$$

Break!

What was the point of all this?

Interpreting the dual

Primal LP

$$\begin{array}{ll} \max & 20x_1 + 30x_2 \quad (P) \\ \text{s.t.} & x_1 + 2x_2 \leq 30 \quad [p_1] \\ & 10x_1 + 5x_2 \leq 120 \quad [p_2] \\ & x_1, x_2 \geq 0 \end{array}$$

Dual LP

$$\begin{array}{ll} \min & 30p_1 + 120p_2 \quad (D) \\ \text{s.t.} & p_1 + 10p_2 \geq 20 \quad [x_1] \\ & 2p_1 + 5p_2 \geq 30 \quad [x_2] \\ & p_1, p_2 \geq 0 \end{array}$$

The Dual LP is from the perspective of the procurement division:

- Decision variables p_1, p_2 : price at which to acquire labor and wood
- Objective: minimize total spend
- Constraints: make sure that supplier does not have incentive to sell directly to your customers
 - $p_1 + 10p_2 \geq 20$: value of resources required by table (1 labor hour and 10 units of wood) should not be less than \$20
 - $2p_1 + 5p_2 \geq 30$: value of resources required by chair (2 labor hour and 5 units of wood) should not be less than \$30

Interpreting the dual (cont.)

Primal LP

$$\begin{array}{ll}\max & 20x_1 + 30x_2 \quad (P) \\ \text{s.t.} & x_1 + 2x_2 \leq 30 \quad [p_1] \\ & 10x_1 + 5x_2 \leq 120 \quad [p_2] \\ & x_1, x_2 \geq 0\end{array}$$

Dual LP

$$\begin{array}{ll}\min & 30p_1 + 120p_2 \quad (D) \\ \text{s.t.} & p_1 + 10p_2 \geq 20 \quad [x_1] \\ & 2p_1 + 5p_2 \geq 30 \quad [x_2] \\ & p_1, p_2 \geq 0\end{array}$$

- (P) has optimal solution $(x_1^*, x_2^*) = (6, 12)$, optimal objective 480
- (D) has optimal solution $(p_1^*, p_2^*) = (\frac{40}{3}, \frac{2}{3})$, optimal objective 480
- One dual variable per primal constraint
 - $p_1^* = \frac{40}{3}$: having 1 more labor hour increases profit by $\$ \frac{40}{3}$
 - $p_2^* = \frac{2}{3}$: having 1 more unit of wood increases profit by $\$ \frac{2}{3}$
 - Does this hold forever?

Julia demo

Summary

- Linear optimization: decision variables, constraints, and objective
- Standard form of linear optimization
- Geometry: feasible region of LP is a polyhedron
- “Corners are optimal”: if an optimal solution exists, and if corners exist, then there is an optimal corner
- Duality: derivation and general transformation
- Interpretation of the dual as shadow prices
- Julia implementation: formulating and solving LPs, plotting, querying solutions and shadow prices

These will all be covered again in the first few weeks of 15.C57 but in greater detail!